

supplier rates. The *reserve factor* is a percentage of the borrow payments not given to the suppliers and instead set aside in a reserve pool that acts as insurance in case a borrower defaults. In an extreme price movement, many positions may become undercollateralized in that they have insufficient funds to repay the suppliers. In the event of such a scenario, the suppliers would be repaid using the assets in the reserve pool.

Here is a concrete example of the rate mechanics. In the DAI market, 100 million DAI is supplied, and 50 million is borrowed. Suppose the base rate is 1 percent and the slope is 10 percent. At 50 million borrowed, utilization is 50 percent. The borrow interest rate is then calculated to be $0.5 \cdot 0.1 + 0.01 = 0.06$, or 6 percent. The maximum supply rate (assuming a reserve factor of zero) would simply be $0.5 \cdot 0.06 = 0.03$, or 3 percent. If the reserve factor is set to 10, then 10 percent of the borrow interest is diverted to a DAI reserve pool, lowering the supply interest rate to 2.7 percent. Another way to think about the supply interest rate is that the 6 percent borrow interest of 50 million is equal to 3 million of borrow payments. Distributing 3 million of payments to 100 million of suppliers implies a 3 percent interest rate to all suppliers.

For a more complicated example involving a kink, suppose 100 million DAI is supplied, and 90 million DAI is borrowed – a 90 percent utilization. The kink is at 80 percent utilization, before which the slope is 10 percent and after which the slope is 40 percent, which implies the borrow rate will be much higher if the 80 percent utilization is exceeded.